

EDUCATION

University of California, San Diego; M.S Electrical & Computer Engineering Fall 2025 - Present

- **Concentration:** Intelligent Systems, Robotics & Control
- **Related Coursework:** Digital Image Processing, Statistical Learning, Linear Systems Theory

Indian Institute of Information Technology, Jabalpur; B.Tech Mechanical Engineering 2019 - 2023

- **GPA:** 3.83/4 (Top 5% in my batch of 101 students)
- **Related Coursework:** Human-centric Robotics, Industrial IoT, Smart Sensors, Data Structure & Algorithms, C++

TECHNICAL SKILLS

Areas	Robotics, SLAM, Computer Vision, RL, AI, Motion Planning, Force Control Loop, Human-Robot Interaction, Embedded Systems, Sensor Fusion, Deep Learning, VR/AR, Real-Time Systems, Linux, Multi-threading
Programming	C/C++ (14/17), Python, Matlab, Lua, Shell scripting, C#, Data Structures and Algorithms, Design Patterns
Frameworks	ROS/ROS2, TensorFlow, PyTorch, Scikit-learn, Unity, Isaac Lab/Sim, OpenGL, Git/GitHub
Hardware	Quadrupeds, COBOTs, Haptic Devices, Humanoids, Nvidia Jetson Orin, 3D/2D LiDAR, Realsense Camera, STM-32

PROFESSIONAL EXPERIENCE

Addverb Robotics, Robotics Engineer Aug 2023 - July 2025

Perception

- Lead Architect for **Object Recognition & 6DOF Pose Estimation** modules, designing and deploying from scratch.
- Built a **visual-inertial SLAM pipeline**, fusing **IMU + vision** and reducing tracking dropouts by 35% in low-texture environments.
- Engineered a **self-calibrating camera system** that preserved ± 2 mm accuracy over long durations by correcting for structural droop.

Deep Learning

- Deployed **Reinforcement Learning-driven grasping system** using vision & policy learning, boosting **pick & place success**.
- Applied **semantic segmentation** to classify SKUs, chargers, and conveyors for improved **robotic perception**.
- Linked **scene understanding with error logs** to generate context-aware diagnostics for rare failures, cutting **debugging time** by 45%.

Systems Optimization

- **Dockerized ROS based robotics stack** into modular containers with **CI/CD pipelines** for automated embedded deployment.
- Built an **auto-update workflow** where code pushes generated **Docker images** directly deployed to robots.
- Performed **time-series analysis** for root-cause diagnostics, optimizing **system performance** under real operational constraints.

Deployment

- Led **on-site deployment** for robotic systems with **Siemens, ISITEC, Adidas**, ensuring successful rollout under real-world constraints.
- Implemented **field debugging and rapid iteration**, reducing downtime and accelerating system stabilization post-deployment.

RESEARCH EXPERIENCE

Graduate Student Researcher, Advisor: Prof. Michael Yip Sept 2025 - Present

Advanced Robotics and Controls Lab (ARClab), UC San Diego

- **Project:** Sim-to-Real Transfer for Dexterous Manipulation of Articulated Tools
- Developing a **Reinforcement Learning (RL)** policy in **NVIDIA Isaac Sim** for a multi-fingered humanoid hand.
- Engineering a hybrid controller fusing **tactile & force feedback** via **cross-attention** for robust sim-to-real.
- Validating the controller in **medical robotics**, successfully manipulating articulated surgical tools.

Research Assistant, Neuro-Biohybrid Systems

Aug 2025 - Present

Sanford Consortium for Regenerative Medicine, UC San Diego

- Designing control framework for first-of-its-kind biohybrid system, enabling an **organoid** to control navigation of **quadruped robot**.
- Formulating a novel RL reward function using **optogenetic feedback** signals to induce goal-directed learning in the organoid.
- Engineering **ROS-based middleware** for a closed-loop neuro-biohybrid control system between organoid and robotic agent.

Research Intern

Jan 2023 - July 2023

Defence Research and Development Organisation (DRDO)

- Problem Statement: Identified key challenges in **early warning missile detection systems**.
- Conducted in-depth research on implementing CNNs for **infrared small target detection (IRSTD)**.
- Gained expertise in low **signal-to-noise ratio (SNR) image processing**, shape detection, and DL-based feature aggregation.
- Built and preprocessed IR image datasets, incorporating **pixel-level annotations** and **multi-directional noise analysis**.

HONORS & ACHIEVEMENTS

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| • Published utility patent for the IoT-Enabled 3D Spatial Monitoring Device | 2023 |
| • Gold Medalist , IIIT Jabalpur — best cross-disciplinary project among 400+ students | 2023 |
| • International Grant Awardee , TIGC La Trobe University — Top 10 of 200+ teams across Asia | 2022 |
| • National Winner , L&T Techgium Hackathon — 1st prize of \$12,000, outperforming 30k+ students from 450 universities | 2022 |